

JONATHAN DARKO ADOO

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EDUCATION

Kwame Nkrumah University of Science and Technology (KNUST)

B.Sc. Computer Engineering

Kumasi, Ghana

Expected Graduation: September, 2026

TECHNICAL SKILLS

Languages: Python, C++, JavaScript (ES6+), TypeScript, SQL

Frameworks & Libraries: React, Next.js, Node.js, Express, TailwindCSS

Databases: PostgreSQL, MySQL, MongoDB, Supabase

Tools: Git, GitHub Actions (CI/CD), Docker, Linux (Ubuntu), REST APIs, CI/CD pipelines

Core Competencies: Full-Stack Development, Algorithms, Robotics (ROS2, Jetson, SLAM)

EXPERIENCE

Mikrobot Academy

Software Engineer & Robotics Trainer | Accra & Kumasi | August 2022 - Present

- Designed and deployed a highly optimized academy website from the ground up, reducing page load times by 55% through performance tuning, improving applicant experience and conversion rates.
- Built a secure, full-stack admin portal to manage 100+ incoming student applications, upload dynamic news items, and maintain gallery assets.
- Instructed 50+ students in Arduino and LEGO EV3 programming; led 3 teams to win local robotics competitions.

Center for Scientific and Industrial Research(CSIR-INSTI)

Software Engineering Intern | Accra | September - December 2024

- Optimized website load times by 40% through code refactoring, image compression, and caching strategies.
- Collaborated on building interactive web dashboards using React and Node.js for ongoing research projects, reducing data analysis time by 30% and enabling real-time insights for cross-functional teams.

A2SV Hackathon

Frontend Developer

- Directed a 4-member team to integrate REST APIs and deliver a working MVP within a 7-day timeline, utilizing rapid prototyping to meet hackathon deadlines.
- Advanced to the quarter-finals out of 200+ competing teams for innovation and rapid prototyping capabilities.

PROJECTS

Absense (Smart Attendance System) | Next.js, Node.js, Python, FastAPI, Mobile & Web

- Spearheaded a 4-person engineering team to launch a full-stack smart attendance app for the KNUST College of Engineering, successfully deploying to university servers.
- Deployed core data models alongside GPS geofencing, face verification, and device binding to ensure 99% attendance tracking accuracy for 500+ users in university pilot.

Snipp (AI-Powered Code Workspace) | Next.js, Supabase, Liveblocks, Gemini AI

- Engineered a real-time collaborative workspace for code snippets handling concurrent user edits with near-zero latency for seamless multi-user sessions.
- Integrated the Gemini AI API to power a context-aware search engine, enabling developers to query and locate relevant snippets in under 2 seconds, improving search efficiency by 40%.

ROS2 Robotics Mapping Project - Python, C++, ROS2 Jazzy

- Programmed a Jetson-powered differential drive robot using RPLidar for real-time SLAM mapping.
- Implemented sensor data visualization and robot navigation in simulated and physical environments.

CERTIFICATIONS & ACHIEVEMENTS

- Google UX Design** - Coursera (2023)
- 1st Place** - World Robofest Unknown Mission Challenge